

Tværfaglig Studieopgave 1

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Harry's Salon

User Stories

Acceptance Criteria

FURPS(+)

Systemudvikling

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Create Booking

Story Point

5

User Story:

As a Salon Owner (Harry) / Assistant (Harriet),
I can create a booking by selecting from available time slots on a requested date,
so that appointments are registered and double bookings are prevented.

Acceptance Criteria:

- Given a customer requests a date
When the Salon Owner (Harry) or Assistant (Harriet) opens the booking flow
Then only unbooked slots between 10:00–18:00 Mon–Fri are shown
- Given an available slot is selected and customer details entered
When the booking is confirmed
Then the appointment is saved and appears in the upcoming list
- Given all slots on a date are fully booked
When the date is entered
Then the system informs the user no slots are available
- Given input is invalid (wrong date format, outside opening hours)
When the booking is submitted
Then the system shows an error and does not crash

Delete Booking

Story Point

3

User Story:

As a Salon Owner (Harry) / Assistant (Harriet),
I can delete an existing appointment,
so that cancelled bookings are removed from the system.

Acceptance Criteria:

Given an existing appointment is selected for deletion
When the Salon Owner (Harry) or Assistant (Harriet) confirms the deletion
Then the appointment is removed from the list and the file is updated

Given a non-existent appointment ID is entered
When deletion is attempted
Then the system shows an error and does not crash

File Persistence – Save on Change, reload on startup

Story Point

3

User Story:

As a Salon Owner (Harry),

I need all appointment and payment data to survive a program restart,
so that no records are lost between sessions.

Acceptance Criteria:

Given an appointment is created, updated, or deleted

When the action completes

Then the change is immediately written to file

Given the program is restarted

When it loads

Then all previously saved appointments and payments are restored correctly

Given the data file is empty or does not exist

When the program starts

Then it initializes with an empty state without crashing

View Upcoming Appointments Sorted by Date and Time

Story Point

3

User Story:

As a Salon Owner (Harry) / Assistant (Harriet),
I can view all upcoming appointments sorted by date and time,
so that today's appointments are shown first.

Acceptance Criteria:

Given appointments exist across multiple dates

When the Salon Owner (Harry) opens the appointment list

Then they are displayed in ascending date and time order

Given no upcoming appointments exist

When the list is opened

Then the system displays a message indicating the list is empty

Register Payment for Completed Appointment

Story Point

5

User Story:

As a Salon Owner (Harry),
I can register that a completed appointment has been paid,
so that the salon's income is tracked.

Acceptance Criteria:

- Given a completed appointment exists
- When the Salon Owner (Harry) marks it as paid
- Then the appointment status updates and the payment is saved to file

- Given an appointment is already marked as paid
- When the Salon Owner (Harry) tries to register payment again
- Then the system prevents a duplicate and informs the user

Accountant Lookup – With Sorting

Story Point

5

User Story:

As an Accountant (Revisor),
I can look up a past date and see all customers served with their totals, sorted by name or payment amount,
so that I can verify the salon’s financial records efficiently.

Acceptance Criteria:

Given a past date is entered
When the Accountant searches for it
Then all appointments from that date are listed with customer name and total amount

Given a list of results is displayed
When the Accountant selects sort by name
Then the list reorders alphabetically by customer name

Given a list of results is displayed
When the Accountant selects sort by amount
Then the list reorders by payment amount descending

Given a date with no appointments is entered
When the Accountant searches for it
Then the system displays a message indicating no records found

Given a future date or today’s date is entered
When the Accountant searches for it
Then the system rejects it with an appropriate message

Password for Financial Access

Story Point

1

User Story:

As a Salon Owner (Harry) / Accountant (Revisor),
I must enter a password to access financial data,
so that sensitive information is protected.

Acceptance Criteria:

Given the financial menu is opened
When the Salon Owner (Harry) or Accountant (Revisor) is prompted
Then access is only granted if “hairyharry” is entered correctly

Given the wrong password is entered
When the prompt appears
Then access is denied and the user is returned to the main menu

Register Closed Days

Story Point

2

User Story:

As a Salon Owner (Harry),
I can register a closed day,
so that bookings cannot be made on holidays or days off.

Acceptance Criteria:

Given a date is registered as closed
When a booking is attempted on that date
Then the system rejects it and informs the user the salon is closed

5 Day Availability View

Story Point

2

User Story:

As a Salon Owner (Harry) / Assistant (Harriet),
I can see available slots across the next 5 working days,
so that I can offer alternatives when the requested day is full.

Acceptance Criteria:

Given a requested date is fully booked
When the availability view is opened
Then free slots for the next 5 working days are also shown

Product Add-ons to Appointment

Story Point

3

User Story:

As a Salon Owner (Harry) / Assistant (Harriet),
I can register product purchases as an add-on to a completed appointment,
so that the total bill reflects both the haircut and any products bought.

Acceptance Criteria:

- Given a completed appointment exists
- When a product purchase is added
- Then the total bill updates to include the product amount

- Given no appointment exists for the visit
- When a product-only purchase is attempted
- Then the system rejects it — products can only be added to an existing appointment

Customer Credit Feature

Story Point

5

User Story:

As a Salon Owner (Harry),
I can register if a customer has a credit deal,
so that I can control whether a customer owes money.

Acceptance Criteria:

- Given a past unpaid appointment exists
- When Harry registers a credit payment
- Then the appointment is saved as unsettled and available for later settlement

- Given an unsettled credit appointment exists
- When Harry settles it
- Then the appointment is marked as completed and no longer appears in the unsettled list

- Given the Accountant looks up a date
- When appointments are listed
- Then cash and credit payments are clearly distinguishable

Add Colors to Text

Story Point

2

User Story:

As Salon Owner (Harry) / Assistant (Harriet) and Accountant,
I can see the program in color,
so that everything is easier to distinguish between.

Acceptance Criteria:

Given menus, responses and error messages get displayed
When the program is being used
Then various colors get displayed at their rightful place

FURPS(+)

Functionality

Booking management, slot availability, opening hours, payment registration, sorting, accounting access

Usability

Simple interface and navigation, role separation

Reliability

Input validation, file persistence, crash prevention, startup recovery

Performance

Response time, file read/write

Supportability

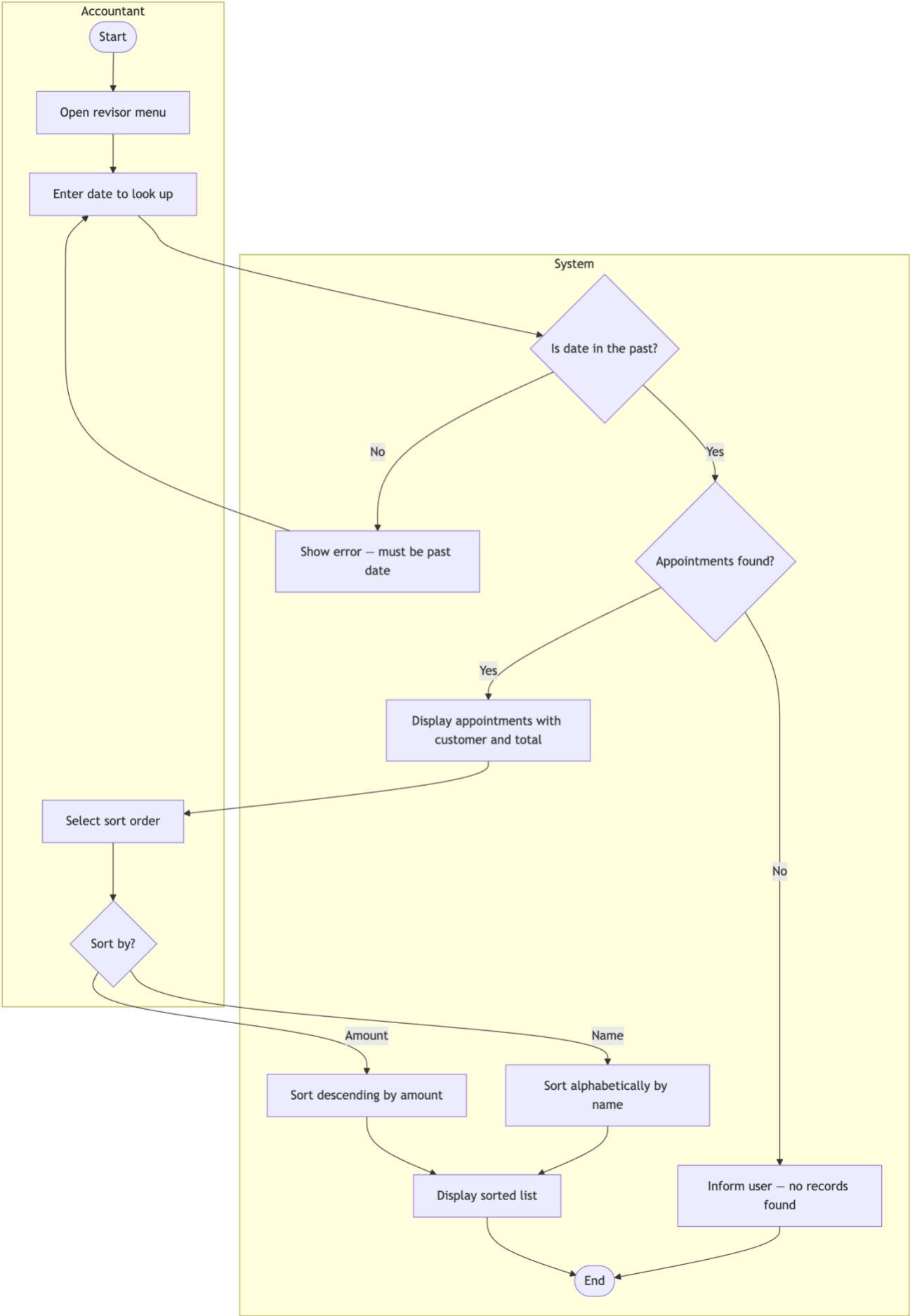
Package structure, JavaDoc, naming conventions, readability

Plus

Only in Java console, offline, single user

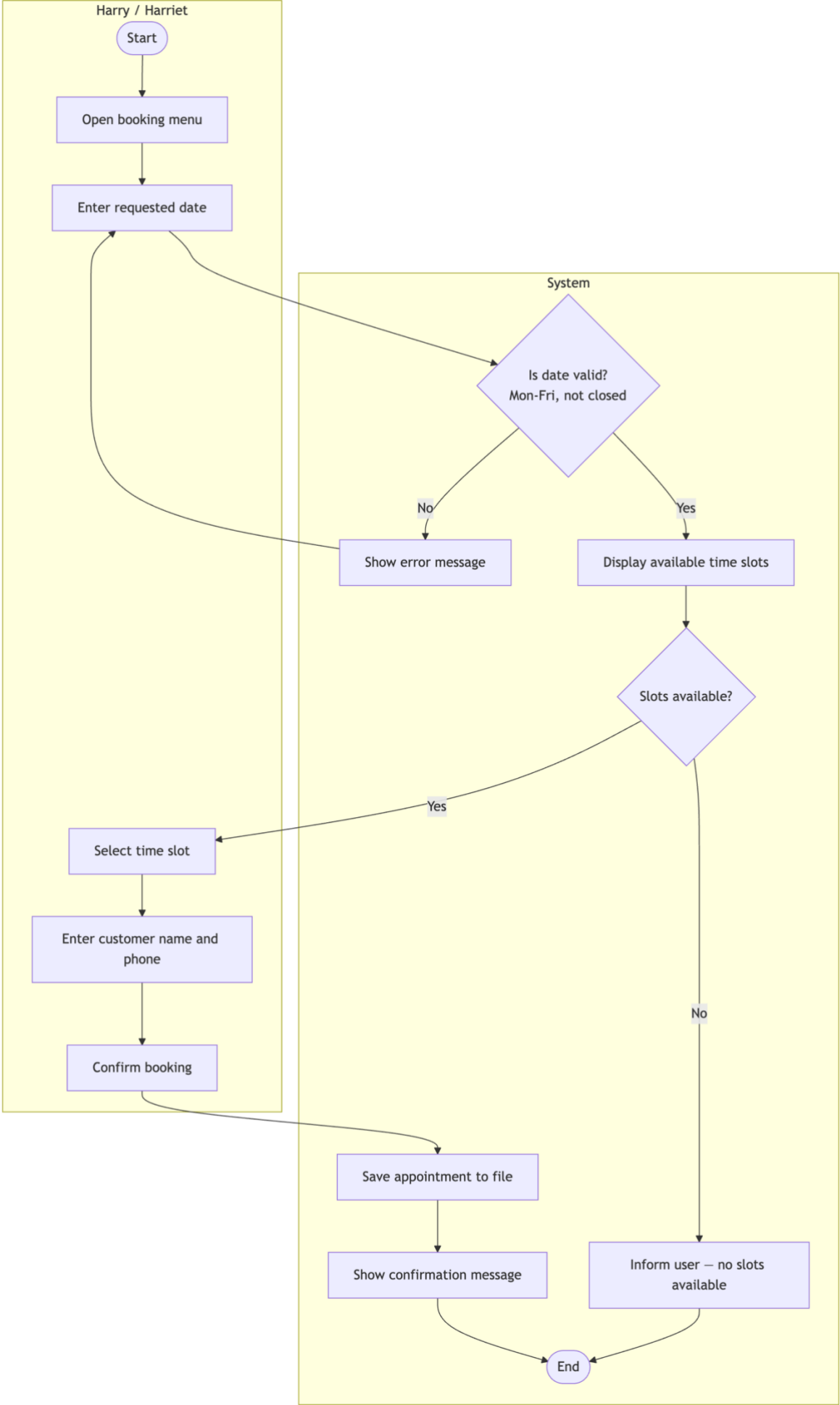
Activity Diagram – Accountant Look-up Flow

Tværfaglig Studieopgave 1 - Gruppe 1 - Harrys Salon - Activity Diagram // Accountant Look-up Flow



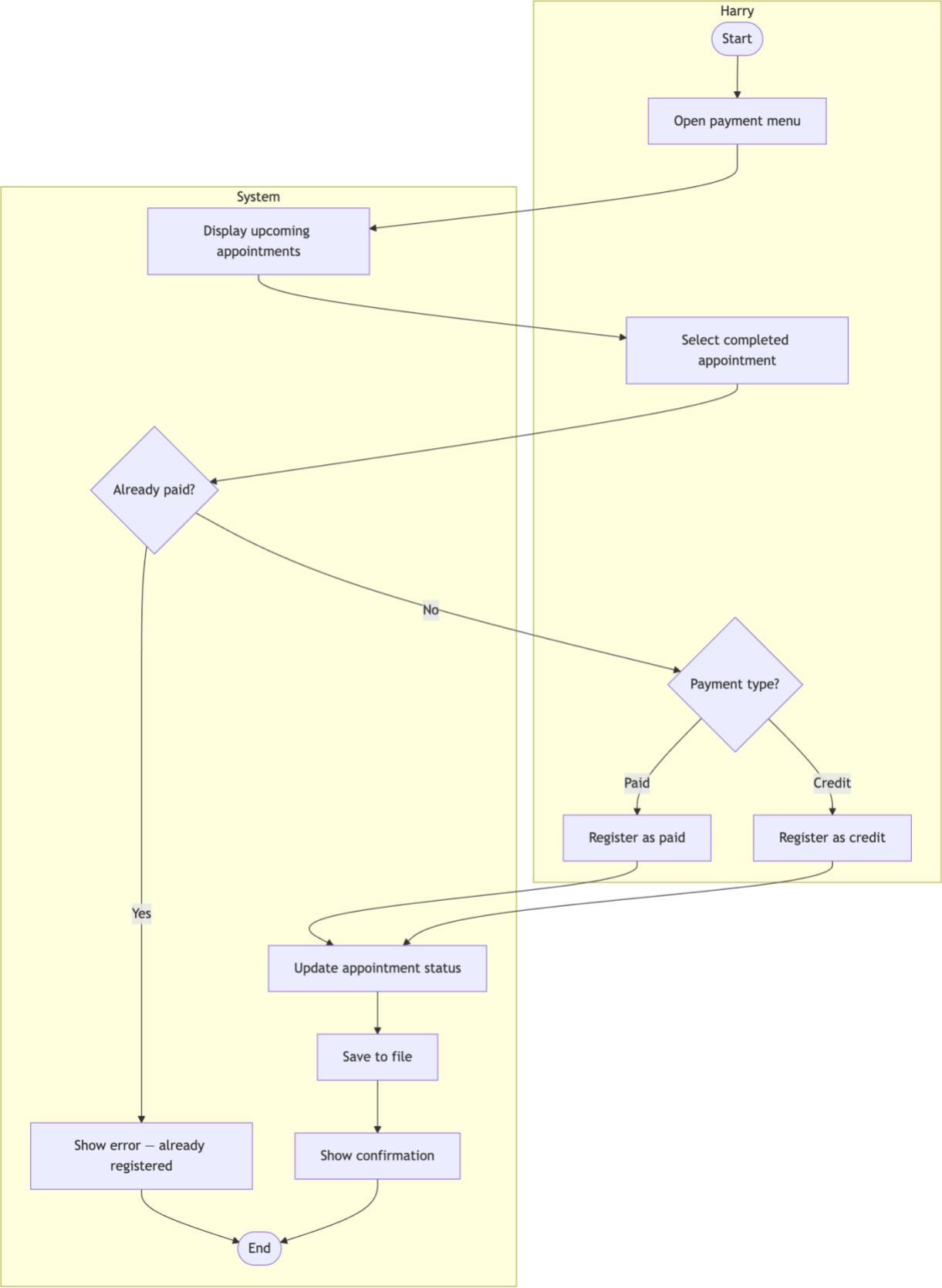
Activity Diagram – Booking Creation Flow

Tværfaglig Studieopgave 1 - Gruppe 1 - Harrys Salon - Activity Diagram // Booking Creation Flow

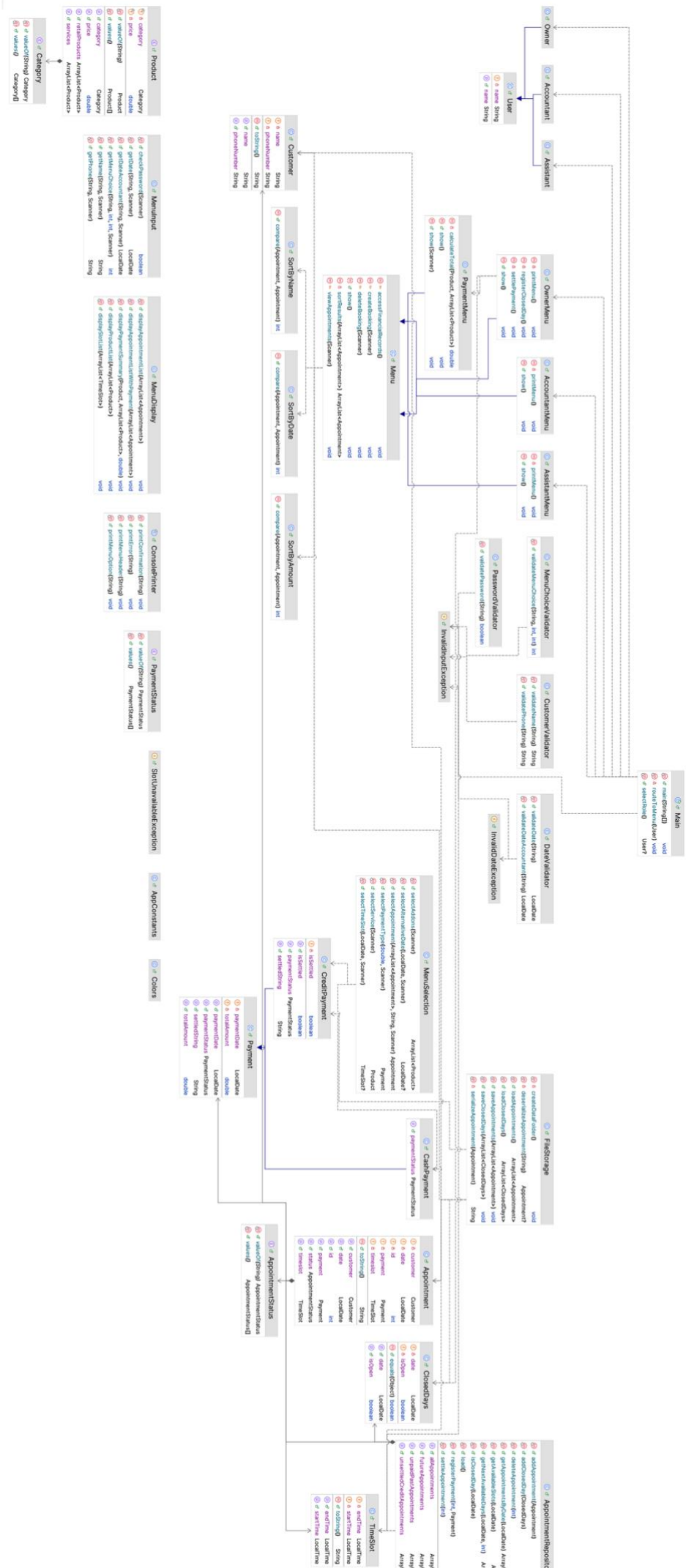


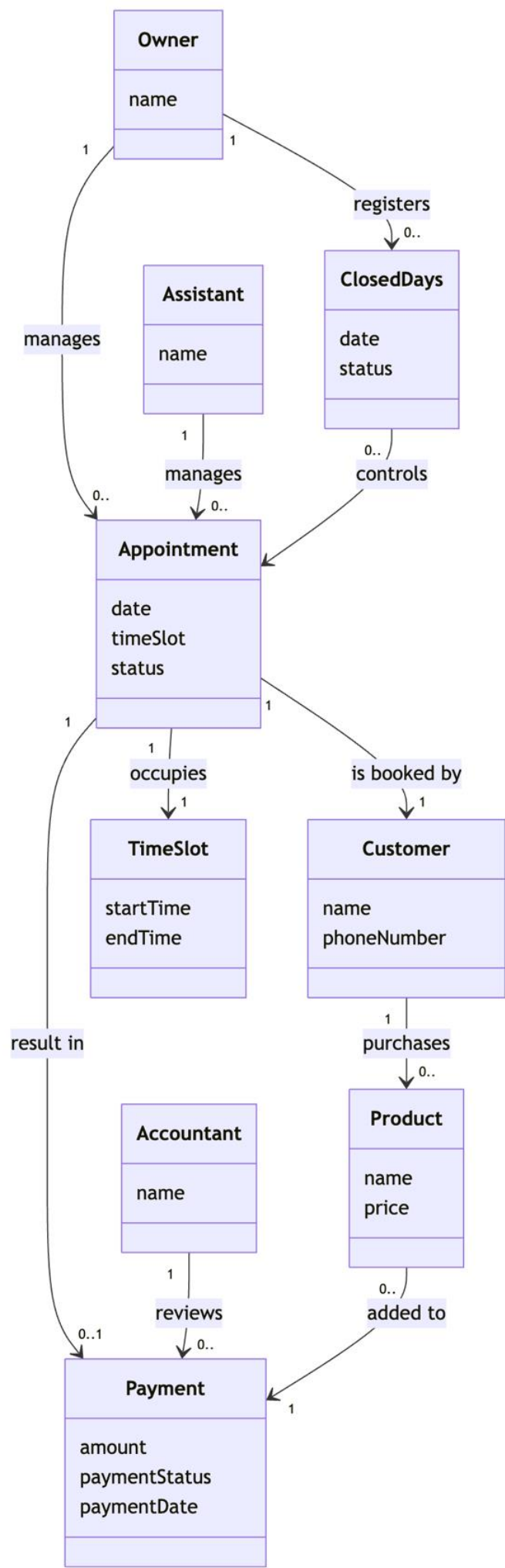
Activity Diagram – Payment Registration Flow

Tværfaglig Studieopgave 1 - Gruppe 1 - Harrys Salon - Activity Diagram // Payment Registration Flow



Class Diagram





SCRUM Process – Sprint 1 – Start of sprint

Trello

q Search

Create

IP PC 👤 🏠 🔍 ⓘ ⚙️ ↺ ➡️ ⚔️ ☰ ☆ 🛖

+ Share

+

Harry's Salon [👥] ▾

Backlog ...

Create Booking
☐ 0/10 P 5

Delete booking
☐ 0/10 P 3View upcoming appointments sorted by date and time
☐ 0/10 P 3Register payment for completed appointment
☐ 0/10 P 5Revisor lookup - with sorting
☐ 0/10 P 5File persistence - save on change, reload on startup
☐ 0/10 P 3Write FURPS+ requirementsDraw domain modelAdd a card

Next Sprint ...

Sprint 2 Goal

Add a card

Sprint Backlog ...

Sprint 1 Goal

Add a card

In Progress ...

Add a card

Code Review ...

Add a card

Done Sprint 1 - 16/03-/20/03 ...

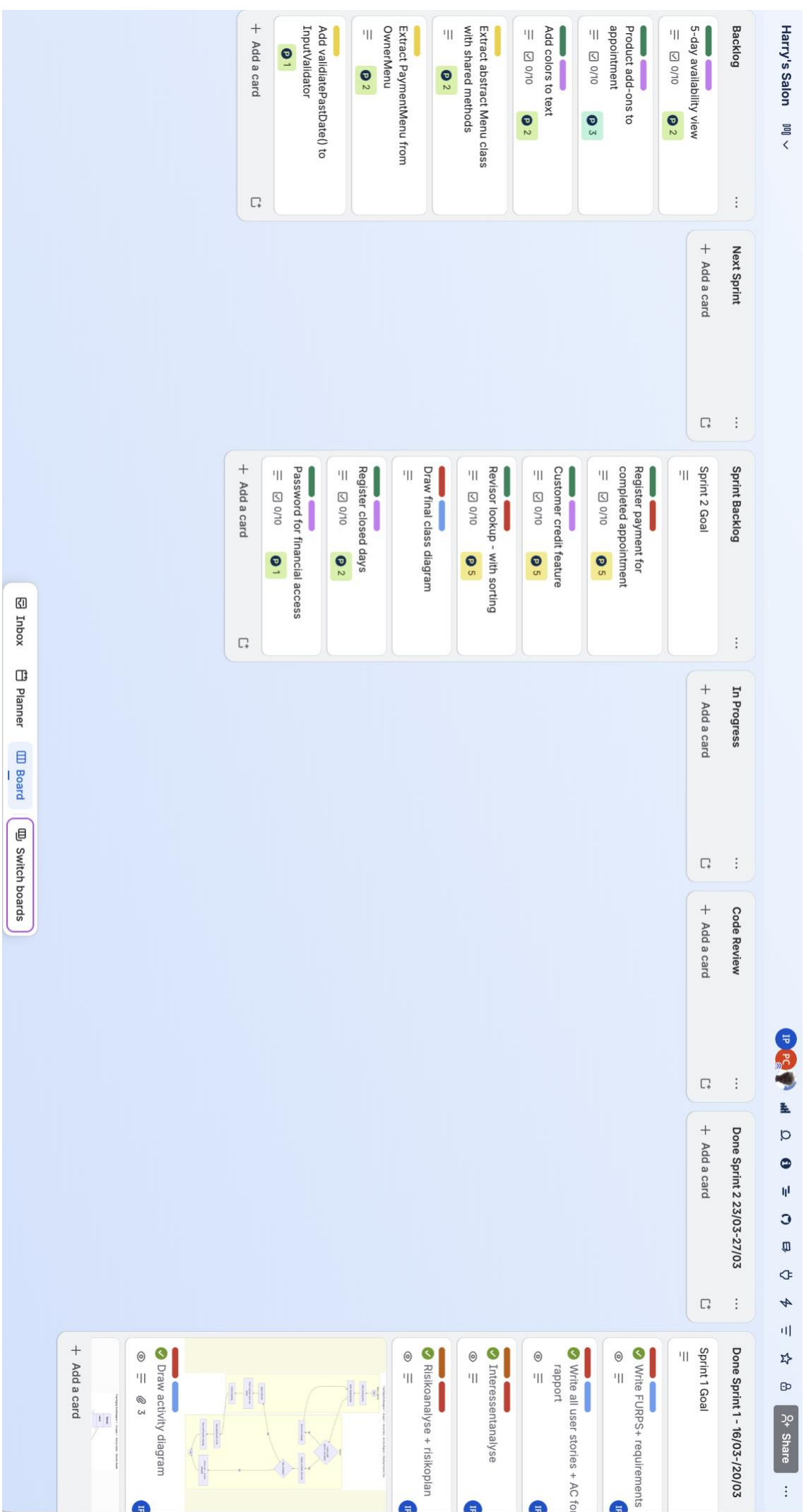
Add a card

Inbox Planner Board Switch boards

SCRUM Process – Sprint 1 – During sprint

[illegible]

SCRUM Process – Sprint 1 – End of sprint/start of sprint 2



SCRUM Process – Sprint 2 – During sprint

Trello

Harry's Salon | Search

Backlog ... + Add a card

Next Sprint ... + Add a card

Sprint Backlog ... Sprint 2 Goal = + Add a card

In Progress ...

- Add colors to text = 0/10 22 hours P 2
- Draw final class diagram @ = 22 hours P 5 IP

+ Add a card

Code Review ...

- Register payment for completed appointment = 8/10 P 5 PC
- Accountant lookup - with sorting = 8/10 P 5
- Product add-ons to appointment = 8/10 P 3 PC
- 5-day availability view = 8/10 P 2
- Customer credit feature = 8/10 P 5 PC

+ Add a card

Done Sprint 2 23/03-27/03 ...

- Segment InputValidator P 1
- ✓ Register closed days = 10/10 P 2
- ✓ Password for financial access Q 5 @ = 10/10 P 1 IP

+ Add a card

Done Sprint 1 -16/03-/20/03 ... + Add another list

- Sprint 1 Goal =
- ✓ Write FURPS+ requirements @ = P 2 IP
- ✓ Write all user stories + AC for rapport @ = P 5 IP
- ✓ Interestanalyse @ = P 2 IP
- Riskanalyse + riskoplan @ = P 3 IP

+ Add a card

Draw activity diagram @ = 3 P 3 IP

SCRUM Process – Sprint 2 – End of sprint/finished project

88Trelio

Harry's Salon

...

Search

Create

jp

...

Backlog

...

Next Sprint

...

Sprint Backlog

...

In Progress

...

Code Review

...

Done Sprint 2 23/03-27/03

...

Done Sprint 1 - 16/03-20/03

...

+ Add another list

+ Add a card

...

+ Add a card

...

+ Add a card

...

+ Add a card

...

+ Add a card

...

✓ Sprint 2 Goal

⌵

jp

pc

✓ Register payment for completed appointment

⌵ 10/10

➡ 5

pc

✓ Add colors to text

⌵ 10/10

➡ 2

jp

✓ Accountant lookup - with sorting

⌵ 10/10

➡ 5

jp

✓ Product add-ons to appointment

⌵ 10/10

➡ 3

pc

✓ Register closed days

⌵ 10/10

➡ 2

jp

✓ Customer credit feature

⌵ 10/10

➡ 5

pc

✓ 5-day availability view

⌵ 10/10

➡ 2

pc

✓ Password for financial access

⌵ 3

⌵ 10/10

➡ 1

jp

+ Add a card

⌵

✓ Sprint 1 Goal

⌵

jp

pc

✓ Write EURLPs+ requirements

⌵

➡ 2

jp

✓ Write all user stories + AC for rapport

⌵

➡ 5

jp

✓ Interestanalyse

⌵

➡ 2

jp

✓ Risikoanalyse + riskopian

⌵

➡ 3

jp

✓ Draw activity diagram

⌵

⌵ 3

➡ 3

jp

+ Add a card

⌵

Inbox

Planner

Board

Switch boards